

Design Calculation Sheet — t8_r3 (VBF-T8R3-CALC-01)

Sheet: Input parameters

Parameter	Value	Unit	Source
Positive particle radius	1.5e-06	m	override
Negative particle radius	2.5e-06	m	override
Electrolyte conductivity	2	S/m	override (estimate)
Electrolyte diffusivity	6e-10	m ² /s	override (estimate)
Cation transference number	0.4	—	override (estimate)
Positive electrode thickness	7.56e-05	m	Chen2020
Negative electrode thickness	8.52e-05	m	Chen2020
Positive porosity	0.335	—	Chen2020
Negative porosity	0.3	—	override 0.25→0.30
Separator thickness/porosity	9e-06 / 0.47	m/—	override thickness / Chen2020 porosity
Collectors Al / Cu	1e-05 / 8e-06	m	override
Heat transfer coefficient	60	W/m ² K	override (cooling design)
Electrode height × width	0.065 × 1.58	m	Chen2020
Nominal capacity	5	Ah	Chen2020
Voltage window	2.5–4.2	V	Chen2020

Sheet: Capacity and energy

Quantity	Value	Formula	Source
1C capacity	5.072795096400077	CC discharge 2.5–4.2 V, 1C (=Nominal capacity)	1c_dfn
Discharge energy	18.56818981379996	$\int V \cdot I \, dt$ (trapezoid, I=1C)	energy
Midpoint voltage	3.95661444268934	voltage at discharge-time midpoint	energy
5C capacity	5.004668867885203	5C CC discharge	5c_dfn
5C retention	0.9865702778802915	5C capacity ÷ 1C capacity	derived

Sheet: Energy density

Quantity	Value	Formula	Source
Stack mass (kg)	0.03732025138660001	$\Sigma \text{ layer thickness} \times (1 - \text{porosity}) \times \text{density} \times \text{area}$ (electrolyte excluded)	energy
ED (Wh/kg)	497.5365685898607	energy_wh ÷ mass_kg	energy
ED (Wh/L)	962.727850372216	energy_wh ÷ volume	energy
Volume (m ³)	1.928706e-05	$\Sigma \text{ layer thickness} \times \text{area}$	energy

Sheet: NP ratio and mass

Quantity	Value	Formula	Source
N/P (full-swing inference)	0.6673617155573662	$(vf_neg \times c_max_neg \times th_neg) / (vf_pos \times c_max_pos \times th_pos)$, F/3600 cancels	parameter set
Positive layer mass (g)	16.8421620276	$layer_kg_m2 \times area$	energy
Negative layer mass (g)	10.149171396	$layer_kg_m2 \times area$	energy
Collectors + separator (g)	10.328917963	$layer_kg_m2 \times area$	energy
Total (g)	37.32025138660001	Σ	energy

Sheet: Process parameters

Parameter	Value	Formula	Source
Positive areal density (g/m ²)	163.993788	$thickness \times (1 - porosity) \times density$	parameter set
Negative areal density (g/m ²)	98.82347999999999	same	parameter set
Positive compaction (g/cm ³)	2.16923	$density \times (1 - porosity) / 1000$	parameter set
Negative compaction (g/cm ³)	1.1599	same	parameter set
Electrolyte fill (g)	6.792495840000001	$pore\ volume \times 1200 \times 1.0$	design-spec section 3
Formation	0.1C CC to 4.2 V, 25 °C, 2 cycles	design recommended; production tuning required	design-spec